There are notes from the book Alessandro de Iaco Veris - Practical Astrodynamics

1 The Two-Body Problem

1.1 Position of the problem

Gravitational Parameter

$$\mu = G(M + m) \approx GM \tag{1}$$

Differentail equation of the motion

$$\ddot{r} = -\frac{\mu}{r^3}\hat{r} \tag{2}$$

Specific angular momentum

$$h = r \times \dot{r} = r \times v = \text{const} \tag{3}$$

Eccentricity vector

$$e = \frac{\vec{v} \times \vec{h}}{\mu} - \frac{\vec{r}}{r} \tag{4}$$

True anomaly

$$\cos\theta = \frac{\vec{e} \cdot \vec{r}}{er} \tag{5}$$

Orbital equation

Table 1: Dependence of orbital type on eccentricity e

Orbital type	Eccentricity e	Description
Circular	$e = 0$	Constant radius orbit
Elliptic	$0 < e < 1$	Closed orbit
Parabolic	$e = 1$	Escape boundary
Hyperbolic	$e > 1$	Open escape orbit

$$r = \frac{h^2/\mu}{1 + e\cos\theta} = \frac{p}{1 + e\cos\theta} \tag{6}$$

Vis-viva equation - specific mechanical energy

$$\epsilon = \frac{v^2}{2} - \frac{\mu}{r} = -\frac{\mu}{2a} \tag{7}$$

$$T = 2\pi\sqrt{\frac{a^3}{\mu}} \tag{8}$$

Table 2: Dependence of orbital type on specific mechanical energy ϵ

Orbital type	Specific energy ϵ	Interpretation
Circular	$\epsilon < 0$	Bound orbit
Elliptic	$\epsilon < 0$	Bound closed orbit
Parabolic	$\epsilon = 0$	Marginal escape
Hyperbolic	$\epsilon > 0$	Unbound escape orbit